

Subject Description Form

Subject Code	AMA573
Subject Title	Statistical Machine Learning
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	To introduce students to the key principles and methods of statistical machine learning. The subject develops students' ability to use Python for data analysis and model building, and to apply these skills in evaluating and interpreting results, with a focus on financial and related applications.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Have a good understanding of the basic theory and methods of modern statistical learning. (b) Recognize a wide variety of algorithms to data. (c) Develop skills to assess statistical uncertainties for conclusions based on data and statistical analysis. (d) Develop and implement new methods that are appropriate for specific data problems in applications.
Subject Synopsis/ Indicative Syllabus	Foundations of Statistical Learning and Computational Analytics: Overview of statistical learning, review of probability and statistics, and introduction to Python for data analysis. Regression and Classification Models: Linear and nonlinear regression methods, logistic regression, and related techniques for prediction and classification problems. Temporal Modeling and Dimensionality Reduction Techniques: Basic concepts and models for time-dependent data, principal component analysis, and other approaches for handling high-dimensional datasets. Modern Machine Learning Algorithms and Applications: Distance-based learning methods, decision trees, ensemble techniques, and clustering, with applications in finance and related fields.
Teaching/Learning Methodology	The subject will mainly be delivered through lectures and exercise-based tutorials. The teaching and learning approach is mainly problem-solving oriented. The approach aims at the development of financial modeling and quantitative skills and how the techniques can be applied to applications in

	financial technology. Students are encouraged to adopt a deep study approach by employing high level cognitive strategies, such as critical and evaluative thinking, relating, integrating and applying theories to practice.					
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)			
			a	b	c	d
	1.Assignments/Project	20%	✓	✓	✓	✓
	2.Tests	40%	✓	✓	✓	
	3. Examination	40%	✓	✓	✓	✓
	Total	100%				
	Continuous Assessment comprises of assignments/project and one midterm test. A written examination is held at the end of the semester.					
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:					
	- Assignments: help students understand the basic theory and methods of modern statistical learning.					
	- Project: train students’ skills to assess statistical uncertainties for conclusions based on data and statistical analysis.					
- Tests: a part of continuous assessment for the theoretical part and data analysis skills covered in the subject.						
- Examination: an overall examination of the theory and statistical learning studied in the whole semester.						
Student Study Effort Required	Class contact:					
	▪ Lecture			26 Hrs.		
	▪ Tutorial			13 Hrs.		
	Other student study effort:					
	▪ Assignments/Projects			58 Hrs.		
	▪ Self-study			40 Hrs.		
	Total student study effort			137 Hrs.		

Reading List and References

References:

- | | | |
|---|--|--|
| M. Anthony
and P.L.
Bartlett | Neural Network Learning:
Theoretical Foundations. | Cambridge
University
Press, 2010 |
| T. Hastie and R.
Tibshirani, J.
Friedman | The Elements of Statistical
Learning, 2nd Ed. | Springer
2009 |
| C. Bishop | Pattern Recognition and
Machine Learning | Springer,
2006 |
| M. Mohri, A.
Rostamizadeh
and A.
Talwalkar | Foundations of Machine
Learning | MIT Press,
2018 |